

K Level	Bloom's Taxonomy Action Verbs
K1	Choose, Define, Find, How, Label, List, Recall, Relate, Select, Show, Spell, What, When, Where, Which, Who, Why
K2	Classify, Compare, Contrast, Demonstrate, Explain, Extend, Illustrate, Infer, Interpret, Outline, Relate, Rephrase, Show, Summarize, Translate
K3	Apply, Build, Choose, Construct, Develop, Experiment with, Identify, Interview, Make use of, Model, Organize, Plan, Select, Solve, Utilize
K4	Analyze, Assume, Categorize, Classify, Compare, Conclusion, Contrast, Discover, Dissect, Distinguish, Divide, Examine, Function, Inference, Inspect
K5	Appraise, Assess, Award, Choose, Compare, Conclude, Criticize, Decide, Deduct, Defend, Determine, Disprove, Estimate, Evaluate, Explain, Importance, Influence, Interpret, Judge, Justify, Measure, Opinion, Perceive, Prioritize, Prove, Recommend
K6	Adapt, Build, Change, Choose, Combine, Compile, Compose, Construct, Create, Delete, Design, Develop, Discuss, Elaborate, Estimate, Formulate, Happen, Imagine, Improve, Invent, Makeup, Maximize, Minimize, Modify, Original, Originate, Plan, Predict, Propose, Solution, Solve, Suppose, Test, Theory

Unit V

Processor

Course Objective – CO5: To Understand the concept of various memories and I/O interfacing.

Course Outcome – CO5: Identify the characteristics of various memory systems and I/O communication

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1.	Define Memory Hierarchy. A structure that uses multiple levels of memory with different speeds and sizes. The faster memories are more expensive per bit than the slower memories.	K1
2.	Define spatial locality Spatial locality is a principle in computer organization where memory locations near a recently accessed address are likely to be accessed soon. This concept is used to optimize performance, particularly in caching, where nearby memory blocks are fetched together to reduce future access times.	K1
3.	Define hit ratio When a processor refers a data item from a cache, if the referenced item is in the cache, then such a reference is called Hit. If the referenced data is not in the cache, then it is called Miss, Hit ratio is defined as the ratio of number of Hits to number of references. Hit ratio = Total Number of references	K1
4.	Define temporal locality. Temporal locality is a principle in computer organization that suggests if a memory location is accessed; it is likely to be accessed again in the near future. This behavior is common in programs where certain variables or instructions are repeatedly used within short time intervals.	K1
5.	How cache memory is used to reduce the execution time If active portions of the program and data are placed in a fast small memory, the average memory access time can be reduced, thus reducing the total execution time of the program. Such a fast small memory is called as cache memory.	K2
6.	Define Hit and Miss? The performance of cache memory is frequently measured in terms of a quantity called hit ratio. When the CPU refers to memory and finds the word in cache, it is said to produce a hit. If the word is not found in cache, then it is in main memory and it counts as a miss	K1

7.	What is cache memory? It is a fast memory that is inserted between the larger slower main memory and the processor. It holds the currently active segments of a program and their data	K1						
8.	What is the necessary of virtual memory? State the advantages of virtual memory? Virtual memory is an important concept related to memory management. It is used to increase the apparent size of main memory at a very low cost. Data are addressed in a virtual address space that can be as large as the addressing capability of CPU	K1						
9.	Define virtual memory. The data is to be stored in physical memory locations that have addresses different from those specified by the program. The memory control circuitry translates the address specified by the program into an address that can be used to access the physical memory	K1						
10.	What is DMA ?(NOV/DEC 2014) Direct memory access (DMA) is a method that allows an input/output (I/O) device to send or receive data directly to or from the main memory, bypassing the CPU to speed up memory operations. The process is managed by a chip known as a DMA controller (DMAC).	K1						
11.	Differentiate programmed I/O and interrupt I/O <table><tr><th>programmed I/O</th><th>interrupt I/O</th></tr><tr><td>Programmed I/O is the process of I/O instruction written in computer program</td><td>Interrupt Initiated I//O is done by using interrupt and some special command.</td></tr><tr><td>In Programmed I/O technique to transfer data, required constant motoring on peripheral by CPU, once data transfer is initiated, CPU have to wait for next transfer.</td><td>In Interrupt Initiated IO once data transfer initiated, CPU execute next program without wasting time and the interface keep monitoring the device.</td></tr></table>	programmed I/O	interrupt I/O	Programmed I/O is the process of I/O instruction written in computer program	Interrupt Initiated I//O is done by using interrupt and some special command.	In Programmed I/O technique to transfer data, required constant motoring on peripheral by CPU, once data transfer is initiated, CPU have to wait for next transfer.	In Interrupt Initiated IO once data transfer initiated, CPU execute next program without wasting time and the interface keep monitoring the device.	K4
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12.	what is the purpose of dirty /modified bit in cache memory. A dirty bit or modified bit is a bit that is associated with a block of computer memory and indicates whether or not the corresponding block of memory has been modified.[1] The dirty bit is set when the processor writes to (modifies) this memory. The bit indicates that its associated block of memory has been modified and has not yet been saved to storage.	K1						
13.	What is flash memory? Flash memory is a type of electrically erasable programmable read-only memory (EEPROM). Unlike disks and DRAM, EEPROM technologies can wear out	K1						

	flash memory bits. To cope with such limits, most flash products include a controller to spread the writes by remapping blocks that have been written many times to less trodden blocks. This technique is called wear leveling.	
14.	Explain the role of a DMA controller in a computer system. A DMA controller plays a crucial role in enhancing the efficiency of data transfers within a computer system. It allows peripherals (such as disk drives, network adapters, or sound cards) to transfer data directly to or from the system's memory without involving the CPU in every data transfer cycle. This frees up the CPU to perform other tasks concurrently, improving overall system performance and efficiency.	K2
15.	Define Interrupt I/O Interrupt I/O is a method of managing input/output operations where the CPU is notified by an interrupt signal from an I/O device when the device is ready for data transfer. Instead of continuously checking the device status (as in polling), the CPU performs other tasks until it receives an interrupt, at which point it pauses its current activity to service the I/O device.	K1
16.	Differentiate between parallel and serial data transmission. • Parallel transmission sends multiple bits of data simultaneously over multiple channels, which is faster but more prone to signal degradation over long distances. • Serial transmission sends data one bit at a time over a single channel, which is slower but more reliable over long distances.	K4
17.	What is a page fault in the context of virtual memory? A page fault occurs when a program tries to access data in a page that is not currently in main memory (RAM). The operating system then fetches the page from secondary storage (like a hard disk) and loads it into RAM, resuming the program's execution afterward.	K4
18.	What is thrashing in a virtual memory system? Thrashing occurs when a computer's virtual memory system spends more time swapping pages in and out of memory than executing the actual processes. This leads to a severe drop in performance as the CPU is overloaded with handling memory page transfers.	K1
19.	Explain the concept of polling in I/O operations. Polling is a method in which the CPU repeatedly checks the status of an I/O device to see if it is ready to perform a data transfer. This method is inefficient because the CPU wastes time in constantly checking the device rather than performing other tasks.	K4
20.	What is interrupt? An interrupt is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention. When an interrupt occurs, the processor temporarily halts its current activities and executes a special routine called an interrupt handler (or interrupt service routine, ISR) to address the event.	K1

S.No.	Question	BTL *
1.	Explain in detail about memory Hierarchy with neat diagram	K2
2.	Discuss the various mapping schemes used in cache memory.	K2
3.	Explain the virtual memory address translation and TLB with necessary diagram.	K2
4.	Explain the typical block diagram of a DMA controller and mention how it is used for direct data transfer between memory and peripherals.	K2
5.	What is virtual memory? Explain how the logical address is translated into physical address in the virtual memory system with a neat diagram.	K2
6	With a neat sketch explain the working principle of DMA.	K2
7	Explain the steps involved in the DMA operation	K2
8	Describe the various mechanisms for accessing I/O devices.	K2
9	Explain the different types of interrupts and the different ways of handling interrupts.	K2